Solutions to Exercise Sheet 1

Anurag Singh, Kiet Vo, Gowtham Reddy, Yuhao Wang, Krikamol Muandet

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1 Introduction & From Precise to Imprecise Probabilities

1.1 False Confidence in Decision-Making

Let $Y = \{\text{"rain"}, \text{"no rain"}\}$ denote tomorrow's weather and a denote the agent's action. The payoffs $r(Y, a)$ are given in the table.

If the agent trusts the first forecaster, we use p to denote *the distribution* of Y that corresponds to the first forecaster's prediction, i.e., $p(\text{"rain"}) = 0.5$.

(a) The agent's expected utility is computed with respect to p as $\mathbb{E}_p[r(Y, \text{"umbrella"})] = -0.5$ and $\mathbb{E}_p[r(Y, \text{"no umbrella"})] = 0$.

(b) The optimal action (i.e., maximising the expected utility) is $\arg\max_a \mathbb{E}[r(Y, a)]$ giving the action of not bringing an umbrella. Because the agent decides not to bring an umbrella, if it rains, their utility will be -2.

If the agent trusts the second forecaster, we use q to denote *any distribution* of Y that is compatible with the second forecaster's prediction, i.e., $q(\text{"rain"}) \in [0.0, 1.0]$. We denote the set of such compatible distributions as $\mathcal{Q}$.

(c) The lower expected utilities of the two actions are $\min_{q \in \mathcal{Q}} \mathbb{E}_q[r(Y, \text{"umbrella"})] = -1$ and $\min_{q \in \mathcal{Q}} \mathbb{E}_q[r(Y, \text{"no umbrella"})] = -2$. The upper expected utilities of the two actions are respectively 0 and 2.

(d) In this case, the probability interval does not lead to a unique expected-utility-maximising action, the agent could seek additional information to reduce uncertainty. Alternatively, the agent can apply a decision criterion for imprecise probabilities. For example, under Γ-maximin the agent chooses the action with the highest lower expected utility, i.e., choosing to bring an umbrella.

1.2 Ambiguity Aversion

Consider the subjective interpretation of precise probabilities,

(a) Let q_1, q_2 denote the player's subjective probability distributions for the outcomes of urns A and B, respectively. Since the player maximises expected monetary payoff and prefers $\mathbf{1}_r^A$ to $\mathbf{1}_r^B$, this implies that $\mathbb{E}_{q_1}[\mathbf{1}_r^A] > \mathbb{E}_{q_2}[\mathbf{1}_r^B]$. By the definition of fair prices, we have $p_r^A = \mathbb{E}_{q_1}[\mathbf{1}_r^A]$ and $p_r^B = \mathbb{E}_{q_2}[\mathbf{1}_r^B]$, hence $p_r^A > p_r^B$.

(b) Using the properties of Bernoulli random variables, we can show that q_1("red ball from A") $= \mathbb{E}_{q_1}[\mathbf{1}_r^A]$. Then, the subjective probability q_1("red ball from A") is equal to the fair price p_r^A for this indicator gamble $\mathbf{1}_r^A$. Similarly, q_2("red ball from B") $= p_r^B$.

(c) Similarly, q_1("blue ball from A") $= p_b^A$ and q_2("blue ball from B") $= p_b^B$.

(d) If $p_r^A + p_b^A = 1$, then $p_r^B + p_b^B < 1$, because we already showed that $p_r^B < p_r^A$ and $p_b^B < p_b^A$. This contradicts the additivity of precise probabilities.

Suppose instead that the player represents their uncertainty about drawing a red ball from urn B using the probability interval $[0.0, 1.0]$. Let p denote any distribution of the outcome of urn B that is compatible with the given probability interval, i.e., p("red ball from B") $\in [0.0, 1.0]$, and $\mathcal{P}$ denote the set of all such distributions.

(e) From the interval $[0.0, 1.0]$, we have the lower probability $\underline{p}$("red ball from B") $= 0.0$ and upper probability $\overline{p}$("red ball from B") $= 1.0$.

(f) The lower prevision can be computed as $\underline{P}(1_r^B) = \min_{p \in \mathcal{P}} \mathbb{E}_p[1_r^B] = 0.0$. The upper prevision can be computed in a similar way, giving the result 1.0.

(g) Use the conjugacy rule $\overline{p}(e) = 1 - \underline{p}(e^c)$ (for any event e), we can compute the lower and upper probabilities of the event "drawing a blue ball from urn B" as 0.0 and 1.0 respectively. Then, the corresponding previsions $\underline{P}(1_b^B)$ and $\overline{P}(1_b^B)$ of the gamble $\mathbf{1}_b^B$ are 0.0 and 1.0.

(h) Since the ball proportion in urn A is known as $50:50$, we have $\underline{P}(1_r^A) = \overline{P}(1_r^A) = p_r^A = 0.5$. Similarly, $\underline{P}(1_b^A) = \overline{P}(1_b^A) = p_b^A = 0.5$. Comparing the respective lower previsions, we have $\underline{P}(1_r^A) > \underline{P}(1_r^B)$ and $\underline{P}(1_b^A) > \underline{P}(1_b^B)$, so the player will always choose urn A.

2 Probability Refresher

2.1 Common Pitfalls

(a) A random variable is a measurable function from the outcome space to another space.

(b) This confuses the random variable with its distribution. The right way to say it would be: "Let X be a random variable with a zero-mean Gaussian distribution".

(c) The probability measure is defined on the σ-algebra, and if some subset is absent from it, its probability is not defined.

(d) The event $X = 0.3$ is not measurable typically under the Borel σ-algebra on reals so its probability is 0.

(e) This is not always true, especially if the random variable is defined on a continuous space. For example, consider someone throwing a dart at numbers between $[0, 1]$. Let us assume that the dart lands on some number $x \in [0, 1]$. The $P(\{X = x\}) = 0$. However, the event that the dart landed at x did happen.

2.2 Kolmogorov's Model of Probability

(a) $X(HH) = 2$, $X(HT) = X(TH) = 1$ and $X(TT) = 0$. The preimage of a set $B \subseteq \mathbb{R}$ is $X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\}$, and $P_X(B) = \mathbb{P}(X^{-1}(B))$.

B	$X^{-1}(B)$	$P_X(B)$
$\{0\}$	$\{TT\}$	$\frac{1}{4}$
$\{1\}$	$\{HT, TH\}$	$\frac{1}{2}$
$\{1, 2\}$	$\{HH, HT, TH\}$	$\frac{3}{4}$
$(-\infty, 1.5]$	$\{HT, TH, TT\}$	$\frac{3}{4}$
$\{5\}$	$\emptyset$	0

The set $(-\infty, 1.5]$ contains values that X never takes. Only the values 0 and 1 in it matter. The set $\{5\}$ contains no value of X at all, so its preimage is empty. X has no inverse function, since it is not one-to-one, but every preimage is still defined.

(b) $P_X(\{0, 1, 2\}) = 1$, so P_X has a PMF $p_X(x) = P_X(\{x\})$. It is $p_X(0) = \frac{1}{4}$, $p_X(1) = \frac{1}{2}$, $p_X(2) = \frac{1}{4}$, and $p_X(x) = 0$ for every other x. The CDF $F_X(t) = P_X\big((-\infty, t]\big)$ is

$$F_X(t) = \begin{cases} 0, & t < 0, \\ \frac{1}{4}, & 0 \le t < 1, \\ \frac{3}{4}, & 1 \le t < 2, \\ 1, & t \ge 2. \end{cases}$$

It is a step function that jumps at 0, 1 and 2, and each jump equals the value of p_X there.

(c) The completed table is

Object	Input	Output
X	an outcome $\omega \in \Omega$	a real number $X(\omega)$
X^{-1}	a set $B \subseteq \mathbb{R}$	a set of outcomes $X^{-1}(B) \subseteq \Omega$
$\mathbb{P}$	an event $A \in \mathcal{F}$	a number $\mathbb{P}(A) \in [0, 1]$
P_X	a Borel set $B \subseteq \mathbb{R}$	a number $P_X(B) \in [0, 1]$
p_X	a real number x	a number $p_X(x) \in [0, 1]$
F_X	a real number t	a number $F_X(t) \in [0, 1]$

Only X and X^{-1} involve outcomes. $P_X = \mathbb{P} \circ X^{-1}$ first sends a set of values back to the outcomes that produce them, then measures that set with $\mathbb{P}$. The last three objects all return numbers in $[0, 1]$, but P_X takes sets as input, while p_X and F_X take numbers.

(d) X_1 is a random variable. It takes the value 0 on $\{2, 4, 6\}$ and 1 on $\{1, 3, 5\}$, so every preimage $X_1^{-1}(B)$ is $\emptyset$, $\{2, 4, 6\}$, $\{1, 3, 5\}$ or Ω, and all of these lie in $\mathcal{F}$.

X_2 is not a random variable, since $X_2^{-1}(\{6\}) = \{6\} \notin \mathcal{F}$. In general, on a finite Ω a function is a random variable exactly when it is constant on each of the smallest nonempty sets of $\mathcal{F}$.

$\mathbb{P}(\{6\})$ is not defined, because $\{6\}$ is not an event. The model only says how likely even and odd are. Both the fair die and the die that shows 1 or 2 with probability $\frac{1}{2}$ each agree with $\mathbb{P}$ on $\mathcal{F}$, and they give $\{6\}$ the probabilities $\frac{1}{6}$ and 0. So $\mathcal{F}$ is part of the model: it fixes which questions the model can answer. Compare the coin in (a) to (c), where $\mathcal{F}$ contains every subset and every question has an answer.

3 Betting interpretation of Probability

(a) For any disjoint events A and B, we assume without loss of generality that $P(A \cup B) \leq P(A) + P(B)$. Then, the bookkeeper can buy the gamble $\mathbf{1}_{A\cup B}$ from You, and then sell You the gambles $\mathbf{1}_A$ and $\mathbf{1}_B$. Your earning is

$$P(A \cup B) - (P(A) + P(B)) - (\mathbf{1}_{A\cup B} - \mathbf{1}_A - \mathbf{1}_B) < 0$$

This guarantees a sure loss. Hence, P is not coherent.

(b) Consider any K finite disjoint sets of events $A_1, A_2, \ldots, A_K$ and buy or sell actions $c_1, c_2, \ldots, c_K \in [+1, -1]$. Then, your earnings can be represented by a random variable

$$W = \sum_{k=1}^{K} c_k[\mathbf{1}_{A_k} - P(A_k)]$$

which has zero expectation. A random variable with zero expectation cannot be negative for all outcomes. Hence, there cannot be finite transactions that result in $W < 0$ all the time. Consequently, a pricing scheme that is finitely additive is coherent.